

Computer Networking: Security

(CSE 3752)

Experiment 6

Aim:

To implement asymmetric key cryptography using RSA algorithm and perform Diffie–Hellman key exchange to establish a shared secret key between communicating parties.

Objectives:

1. Understand the working of the RSA public key cryptosystem and the concept of Diffie–Hellman key exchange.
2. Perform encryption and decryption using RSA.
3. Compute shared secret key using Diffie–Hellman algorithm.
4. Analyze how asymmetric cryptography supports secure communication.

Exercises:

1. Given the following values for the RSA algorithm:

- Two prime numbers: $p=61$, $q=53$
- Public key: $e=17$

Calculate the private key d , where d is the modular inverse of e modulo $\phi(n)$.

2. Given the following parameters for Diffie-Helman algorithm used by A and B for key exchange.

- The shared prime $q=157$ and the primitive root $p=5$.

Calculate: (a) The value of Y_A and Y_B transmitted by both A and B.

(b) The value of secured key (K) shared by both A and B.

3. Given a scenario, where B has received a document from A through internet. Explain, how B confirms that the document has been transmitted by A only (not any adversary) using the concept of digital signature as an use of public key cryptosystem.
4. Transform the plaintext “AES USES A MATRIX” into a state matrix form.
5. Compute the output of S-box with given input state matrix as $\begin{bmatrix} 4 & 5 \\ E & 2 \end{bmatrix}$ and S-box as in Fig. 1.

	00	01	10	11
00	1001	0100	1010	1011
01	1101	0001	1000	0101
10	0110	0010	0000	0011
11	1100	1110	1111	0111

Fig. 1.

$$\begin{pmatrix} 63 & EB & 9F & A0 \\ C0 & 2F & 93 & 92 \\ AB & 30 & AF & C7 \\ 20 & CB & 2B & A2 \end{pmatrix}$$

Fig. 2.

6. Perform ShiftRow transformation on the given current matrix in Fig. 2.
7. Find the output of Mix Column if the input state matrix is $\begin{bmatrix} D & 1 \\ A & F \end{bmatrix}$ with the predefined matrix as $\begin{bmatrix} 1 & 4 \\ 4 & 1 \end{bmatrix}$ and the irreducible polynomial for $GF(16)$ is x^4+x+1 .